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SUBJECT: DATA STRUCTURE

SUBJECT CODE: CSA 0302

EXPERIMENT NO: 01

Q1 Reverse a list

1 mark

Write a C program that takes an array as input and returns it reversed.

Your Code

```
1 #include<stdio.h>
2
3 int main(){
4     int n,a[100],i;
5
6     scanf("%d",&n);
7     for(i=0;i<n;i++){
8         scanf("%d",&a[i]);
9     }
10    for(i=n-1;i>=0;i--){
11        printf("%d",a[i]);
12    }
13 }
```

Input

5
1 2 3 4 5

Output

54321

EXPERIMENT NO: 02

Q2 Reverse a string

1 mark

Write a C program for palindrom

Your Code

```
1 #include<stdio.h>
2 #include<string.h>
3
4 int main(){
5     char s[100];
6     int i,n,f=1;
7
8     scanf("%99s",s);
9     n = strlen(s);
10
11    for(i=0;i<n/2;i++){
12        if(s[i]!=s[n-i-1]){
13            f = 0;
14            break;
15        }
16    }
```

Input

madam

Output

palindrome

```

6     int i,n,f = 1;
7
8     scanf("%99s",s);
9     n = strlen(s);
10
11    for(i=0;i<n/2;i++){
12        if(s[i]!=s[n-i-1]){
13            f = 0;
14            break;
15        }
16    }
17    if(f)
18        printf("palindrome");
19    else
20        printf("not palindrome");
21
22    return 0;
23 }

```

Output

palindrome

✓ Submitted

EXPERIMENT NO: 03

QUESTIONS

Q1
Reverse a list
1 mark

Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q3 Searching 1 mark

Write a C program for Searching an Element in a Sorted Array

Your Code

```

1 #include<stdio.h>
2 int main(){
3     int a[100],n,key;
4     int low = 0,high,mid;
5
6     scanf("%d",&n);
7
8     for(int i=0;i<n;i++){
9         scanf("%d",&a[i]);
10
11     scanf("%d",&key);
12
13     high = n-1;

```

Input

5
10 20 30 40 50
30

Output

Element found

EXPERIMENT NO: 04

QUESTIONS

Q1
Reverse a list
1 mark

Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6

Q4 Searching 1 mark

Write a C program for Count the Occurrences of an Element

Your Code

```

1 #include<stdio.h>
2
3 int main(){
4     int a[100],n,key,count = 0;
5
6     scanf("%d",&n);
7
8     for (int i=0;i<n;i++){
9         scanf("%d",&a[i]);
10
11     scanf("%d",&key);
12
13     for(int i=0;i<n;i++){
14         if(a[i] == key)

```

Input

5
10 20 30 20 40
20

Output

2

EXPERIMENT NO: 05

QUESTIONS

Q1

Reverse a list

1 mark

✓

Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q5

Array

1 mark

✓

Q6

Merge Sort

1 mark

✓

Q5 Array

1 mark

Write a program for binary search

Your Code

```
1 #include<stdio.h>
2
3 int main() {
4     int a[100],n,key;
5     int low = 0,high,mid;
6
7     scanf("%d",&n);
8
9     for(int i=0;i<n;i++)
10         scanf("%d",&a[i]);
11
12     scanf("%d",&key);
13
14     high=n-1;
15
```

Input

5
10 20 30 40 50
30

Output

Found

EXPERIMENT NO: 06

QUESTIONS

Q1

Reverse a list

1 mark

✓

Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q6 Merging

Write a C program for Merge Two Arrays

Your Code

```
1 #include<stdio.h>
2
3 int main() {
4     int a[10],b[10],n1,n2,i;
5
6     scanf("%d",&n1);
7     for(i = 0;i<n1;i++)
8         scanf("%d",&a[i]);
9
```

Input

3
10 20 30
2
40 50

Output

1020304050

EXPERIMENT NO: 07

QUESTIONS

Q1

Reverse a list

1 mark

✓

Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q5

Array

1 mark

✓

Q6

Merge Sort

1 mark

✓

Q7 Largest and Smallest element

1 mark

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
1 #include<stdio.h>
2
3 int main(){
4     int n,i;
5     int a[100],largest,smallest;
6
7     scanf("%d",&n);
8
9     for(i=0;i<n;i++){
10         scanf("%d",&a[i]);
11     }
12     largest=smallest=a[0];
13
14     for(i=1;i<n;i++){
15         if(a[i]>largest)

```

Input

5
10 25 5 40 15

Output

largest element = 40
smallest element = 5

EXPERIMENT NO: 08

QUESTIONS

Q1
Reverse a list
1 mark

Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6

Q8 Second Largest Element 1 mark

Write a C program for Find the Second Largest Element

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int n,i;
4     int a[100],largest,second;
5
6     scanf("%d",&a[i]);
7
8     for(i=0;i<n;i++)
9         scanf("%d",&a[i]);
10
11     largest=second= -99999;
12
13     for(i=0;i<n;i++) {
14         if(a[i]>largest){
```

Input

5
10 20 30 40 50

Output

second largest = -99999

EXPERIMENT NO: 09

QUESTIONS

Q1
Reverse a list
1 mark

Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q9 Occurrence of an elemnet 1 mark

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int n,a[100],i,key;
4     int first= -1,last= -1;
5
6     scanf("%d",&n);
7
8     for(i=0;i<n;i++)
9         scanf("%d",&a[i]);
10
11     scanf("%d",&key);
12
13     for(i=0;i<n;i++) {
14         if(a[i] == key){
```

Input

6
2 5 3 5 7 5
5

Output

first occurrence = 1
last occurrence = 5

EXPERIMENT NO: 10

Q1Reverse a list
1 mark**Q2**Reverse a string
1 mark**Q3**Searching
1 mark**Q4**Searching
1 mark**Q5**Array
1 mark**Q6**

Merging

**Q10 Missing Element**

1 mark

Write a C program for Find Missing Element in an Array

Your Code

```
1 #include<stdio.h>
2
3 int main() {
4     int n,i;
5     int a[100];
6     int sum = 0,total,missing;
7
8     scanf("%d",&n);
9
10    for(i=0;i<n-1;i++){
11        scanf("%d",&a[i]);
12        sum += a[i];
13    }
14    total = n*(n+1)/2;
15    missing = total-sum;
```

Input

```
5
1 2 4 5
```

Output

```
missing element = 3
```